

Abdul Basit Tonmoy

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EDUCATION

Wabash College

Crawfordsville, IN

Bachelor of Arts in Computer Science and Mathematics with Minor in Physics

May 2027

Relevant Coursework: Data Structures & Algorithms, Machine Organization, OOP, Machine Learning, Data Science, Numerical Methods, Functional Programming, Stochastic Simulation, Multi-variable Calc, Linear Algebra, Autonomous Robotics, Differential Eq, Compiler Design, Physics Informed Machine Learning

ACHIEVEMENTS

- **Published Research:** *Advanced Facial Emotion Classification with 135 Classes for Enhanced Cybersecurity Applications*, 2025 IEEE ICAIC || *AdaFrame: Hierarchical Multimodal Deduplication with Adaptive Information Budgeting for Cost-Efficient Video Advertisement Analysis*, ACM MM 2026 (in review)
- **J. Crawford Polley Mathematical Writing Prize** (May 2025) | **Dean's List** || **Hackathon Wins:** CalHacks-12, UAP CyberSiege-25

TECHNICAL SKILLS

Languages: Python, JavaScript, C/C++, SQL, HTML/CSS
Backend & Databases: Node.js, FastAPI, Django, RESTful APIs, MongoDB, SQLite, ChromaDB, Neo4j
Data Processing: pandas, NumPy, ETL pipelines, Apache Airflow, Databricks
AI/ML & Cloud: PyTorch, LangChain, Transformers, scikit-learn, AWS (S3, SageMaker, Bedrock), Docker, Git

EXPERIENCE

Skop Intelligence

Berkeley, CA

Co-Founder & Chief Product Officer

Jan. 2026 – Present

- Built Rust-based OCR engine with native TensorRT inference support, achieving 200 docs/sec throughput to process a corpus of 20M+ documents
- Trained neural classification head distinguishing agendas, board minutes, and other document types at 98.2% accuracy, enabling structured downstream analytics
- Closed \$150K licensing deal with American Enterprise Institute for non-commercial access to the largest U.S. public school board minutes and agenda dataset

Compass Group (E15)

Crawfordsville, IN

Data Analyst

Aug. 2025 – Dec. 2025

- Deployed OCR pipeline to automate receipt-to-Excel data ingestion, eliminating manual entry and accelerating financial processing by 10 hours/week

Department of Mathematics and Computer Science, Wabash College

Crawfordsville, IN

Robotics Research Intern

May 2025 – Aug. 2025

- Architected autonomous navigation system fusing GPS, computer vision, and LiDAR data streams, achieving 97% obstacle detection and avoidance and 2-meter precision in real-world testing
- Implemented real-time control algorithms with emergency protocols, enabling safe autonomous operation under varying environmental conditions

Motorpool, Wabash College

Crawfordsville, IN

Software Engineering Intern

May 2024 – Aug. 2024

- Architected full-stack data platform processing 20+ years of fleet records with React/Redux frontend and Node.js/MongoDB backend, reducing data retrieval time by 30% through query optimization
- Engineered real-time dashboard with advanced filtering and search capabilities, serving structured data views with drill-down functionality to transaction-level details
- Built RAG-based automation system using LangChain and GPT-3.5, reducing manual support workload by 25% and saving 5 analyst hours weekly through intelligent query resolution

PROJECTS

SemanticSentry – Universal Semantic Drift Detection Library

2025

Python, PyTorch, FAISS, HuggingFace Transformers, CLIP, W&B, MLflow

- Designed model-agnostic drift monitoring framework quantifying geometric transformations in embedding spaces across fine-tuning, quantization, and LoRA updates without requiring labeled evaluation data
- Implemented multi-metric evaluation suite (CKA for global structure, NPS for local neighborhoods, Isotropy for spectral geometry) with FAISS-accelerated nearest-neighbor search and a 4-tier severity classification predicting downstream degradation
- Architected 5-layer system (encoder adapters, frozen snapshots, metrics registry, transfer functions, MLOps integrations) supporting dual-encoder and multi-modal models with W&B and MLflow hooks for production drift alerting

AdLovin – AI Ad Intelligence System (Winner, CalHacks-12)

Oct 2025

Python, Claude API, ChromaDB, OpenCV, PyTorch, HuBERT, FFmpeg

- Architected multi-agent advertisement analysis pipeline combining computer vision, vector embeddings, and LLMs to extract marketing and business insight signals, achieving 70-90% frame deduplication efficiency and reducing API costs by 70%
- Built scalable data processing infrastructure with parallel visual and audio track analysis, ChromaDB vector database for similarity search and context analysis, and automated frame-to-video consolidation processing 30s videos in 60 seconds
- Integrated 6+ ML models (CLIP embeddings, HuBERT emotion recognition, Haar Cascade detection, EasyOCR) with Claude Vision API, designing JSON schemas for downstream analytics with 85-90% promotional content extraction accuracy